



# Multirate digital signal processing



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# Contents

- Sampling rate conversion (resampling after reconstruction)
  - Upsampling (Interpolation)
  - Downsampling (Decimation)
  - Rational factor
    - several methods for implementing the rate converter: single-stage and multistage
  - Arbitrary factor
- Applications of sampling rate conversion in multirate signal processing systems:
  - narrowband filters
  - digital filter banks
  - subband coding
  - transmultiplexers
  - quadrature mirror filters.





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**Multirate DSP**

Prof. David Kovilpillai  
Department of Electrical Engineering  
Indian Institute of Technology Madras

**Multirate DSP**

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In this chapter we will see Polyphase filter structures, interchange of filters and down samplers/up samplers, sampling rate conversion with cascade integrator comb filters, polyphase structures for decimation and interpolation filters, structures for rational sampling rate conversion

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# What is multirate DSP?

- In many practical applications of digital signal processing, one is faced with the problem of **changing the sampling rate of a signal**, either increasing it or decreasing it by some amount.
- For example, in telecommunication systems that transmit and receive different types of signals (e.g., teletype, facsimile, speech, video, etc.), there is a requirement to **process the various signals at different rates** commensurate with the corresponding bandwidths of the signals.
- The process of converting a signal from a given rate to a different rate is called **sampling rate conversion**.
- In turn, systems that employ multiple sampling rates in the processing of digital signals are called **multirate digital signal processing systems**.



# What is multirate DSP?

EE6130 Multirate DSP

\* Welcome  
 \* Course outline  
 \* Moodle Access

- Theory & applications  
 - Emphasis  
 - Sampling & Reconstruction

DSP

Reading D&S  
 Ch 4 Sec 1-3

$x[n] \xrightarrow{LTI} h[n] \rightarrow y[n]$   
 $\Rightarrow$  underlying sampling rate is same

Multirate DSP  $\Rightarrow$  multiple sampling rates

Wide range Applications
 

- Communications
- Speech, image processing

input  
 output  
 Imp. response





# Example of multirate DSP

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Hi-fidelity audio

Human ear 20 Hz - 20,000 Hz

Most sensitive 2-5 kHz

Speech  $\leq 4$  kHz

Music  $\sim 18$  kHz

Broadcast audio 32 kHz

CD 44.1 kHz

DAT 48 kHz

Task

CD  $\longrightarrow$  Broadcast audio

44.1 kHz 32 kHz



# Two methods for multirate DSP

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NPTEL

Hi-fidelity audio

Human ear 20 Hz - 20,000 Hz

Most sensitive 2-5 kHz

Speech  $\leq 4$  kHz

Music  $\sim 18$  kHz

Broadcast audio 32 kHz

CD 44.1 kHz

DAT 48 kHz

Task

CD  $\longrightarrow$  Broadcast audio

44.1 kHz 32 kHz

Opt 1

CD data  $\longrightarrow$  D/A  $\longrightarrow$  analog CT signal  $\longrightarrow$  A/D  $\longrightarrow$  Broadcast audio

44.1 kHz 32 kHz

- \* Complexity
- \* Analogue components
  - Filters
- \* Loss of quality



# Two methods for multirate DSP

**Hi-fidelity audio**

Human ear 20 Hz - 20,000 Hz  
Most sensitive 2-5 kHz  
Speech  $\leq 4$  kHz  
Music  $\sim 18$  kHz

Broadcast audio 32 kHz  
CD 44.1 kHz  
DAT 48 kHz

**Task**

CD 44.1 kHz  $\rightarrow$  Broadcast audio 32 kHz

**Opt 1**

CD data  $\rightarrow$  D/A  $\rightarrow$  analog CT signal  $\rightarrow$  A/D  $\rightarrow$  Broadcast audio

44.1 kHz  $\rightarrow$  32 kHz

**Opt 2**

CD data  $\rightarrow$  DT proc.  $\rightarrow$  Broadcast audio

Several advantages

- \* Complexity
- \* Analogue components
- \* Loss of quality (Filters)



# Sampling & Reconstruction

\* Time  $\xleftrightarrow{F}$  Frequency

$x_c(t)$   $\xleftrightarrow{F}$   $x_c(j\omega)$

CT CT Fourier FT (CTFT)

$x[n]$   $\xleftrightarrow{F}$   $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$  (freq.)

DT DTFT

$\rightarrow X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$

DT, CF DF7

DT, DF

$k = 0, 1, \dots, N-1$



# Sampling & Reconstruction

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\* Time  $\xrightleftharpoons{F}$  Frequency

$x_c(t) \xrightleftharpoons{F} x_c(j\Omega)$

CT  $\uparrow$  Reconstruction  
Sampling  $\downarrow$

CT Fourier FT (CTFT)

$x[n] \xrightleftharpoons{F} X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$  *freq.*

DT  $\xrightarrow{DTFT}$   $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$

DT, CF  $\xrightarrow{DT, DF}$   $k = 0, 1, \dots, N-1$

2/10



# Nyquist Sampling Theorem

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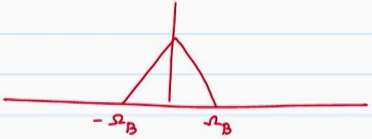
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Bandlimited signal  $x_c(t) \xrightleftharpoons{F} x_c(j\Omega)$

$|x_c(j\Omega)| = 0 \quad |\Omega| \geq \Omega_B$



Nyquist Sampling Theorem

If  $x_c(t)$  is a BL signal with  $|x_c(j\Omega)| = 0 \quad |\Omega| \geq \Omega_B$

Then  $x_c(t)$  is uniquely determined by its samples  $x[n] = x_c(nT_s) \quad n = 0, \pm 1, \pm 2, \dots$

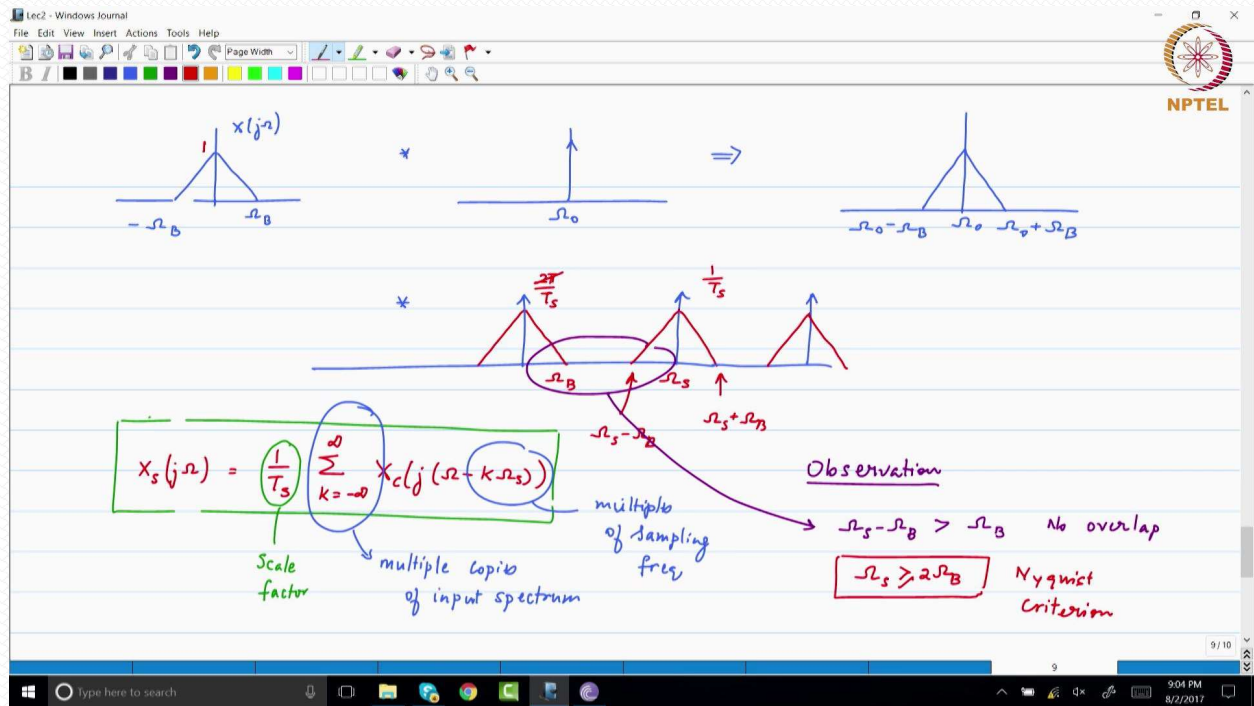
if  $\Omega_s \geq 2\Omega_B$  &  $T_s = \frac{2\pi}{\Omega_s}$

Windows taskbar: Type here to search, [taskbar icons]

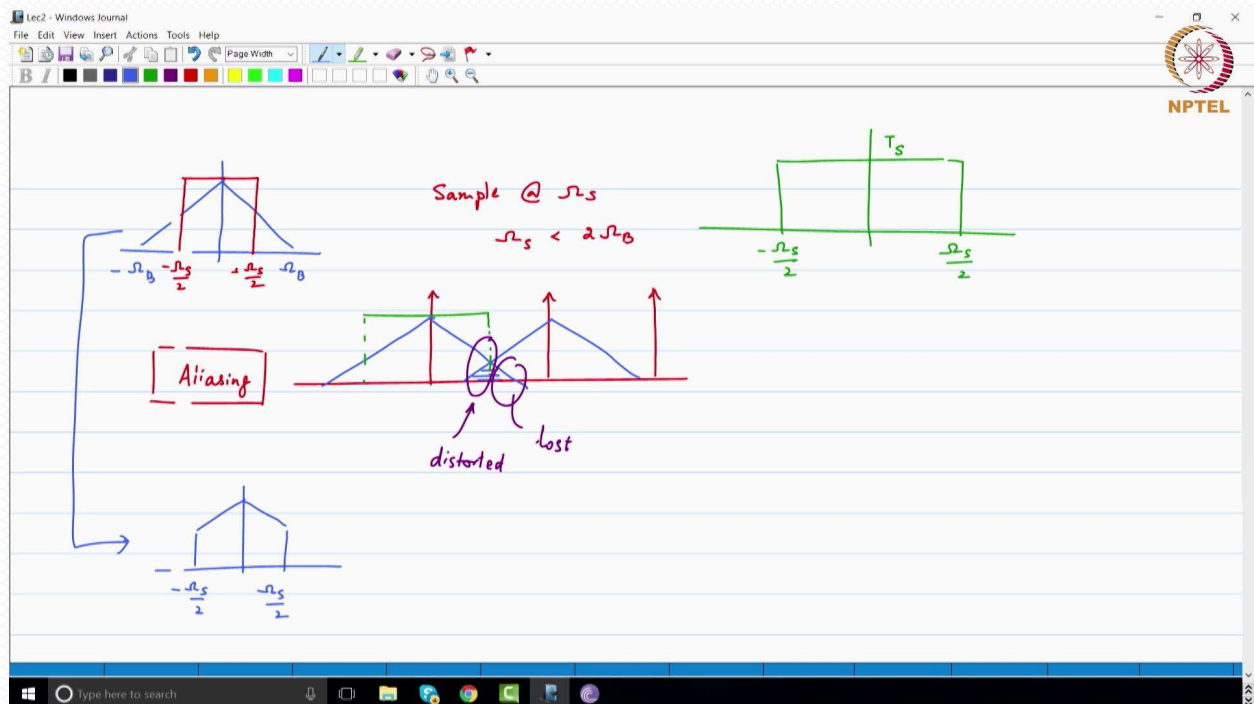




# Nyquist Sampling Theorem



# Nyquist Sampling Theorem







# Upsampling (Interpolation)

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Upsampler / Expander  $x[n] \rightarrow \uparrow L \rightarrow x_E[n]$

(Increase in Sampling rate by integer factor)

$$x_E[n] = \begin{cases} x\left[\frac{n}{L}\right] & \text{if } n = 0, \pm L, \pm 2L, \dots \\ 0 & \text{otherwise} \end{cases}$$

$$X_E(z) = \sum_{n=-\infty}^{\infty} x_E[n] z^{-n} = \sum_{n=-\infty}^{\infty} x\left[\frac{n}{L}\right] z^{-n} \quad n = kL$$

$$= \sum_{k=-\infty}^{\infty} x[k] z^{-kL} \quad \text{mult of } L$$

$$= \sum_{k=-\infty}^{\infty} x[k] z^{-kL} = \sum_{k=-\infty}^{\infty} x[k] z^{-kL}$$

$$X_E(z) = X(z^L)$$

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# Upsampling (Interpolation)

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Expansion

$$y_L[n] = \begin{cases} x\left[\frac{n}{L}\right] & \text{if } n = 0, \pm L, \pm 2L, \dots \\ 0 & \text{otherwise} \end{cases}$$

$M=1$   
 $L=3$

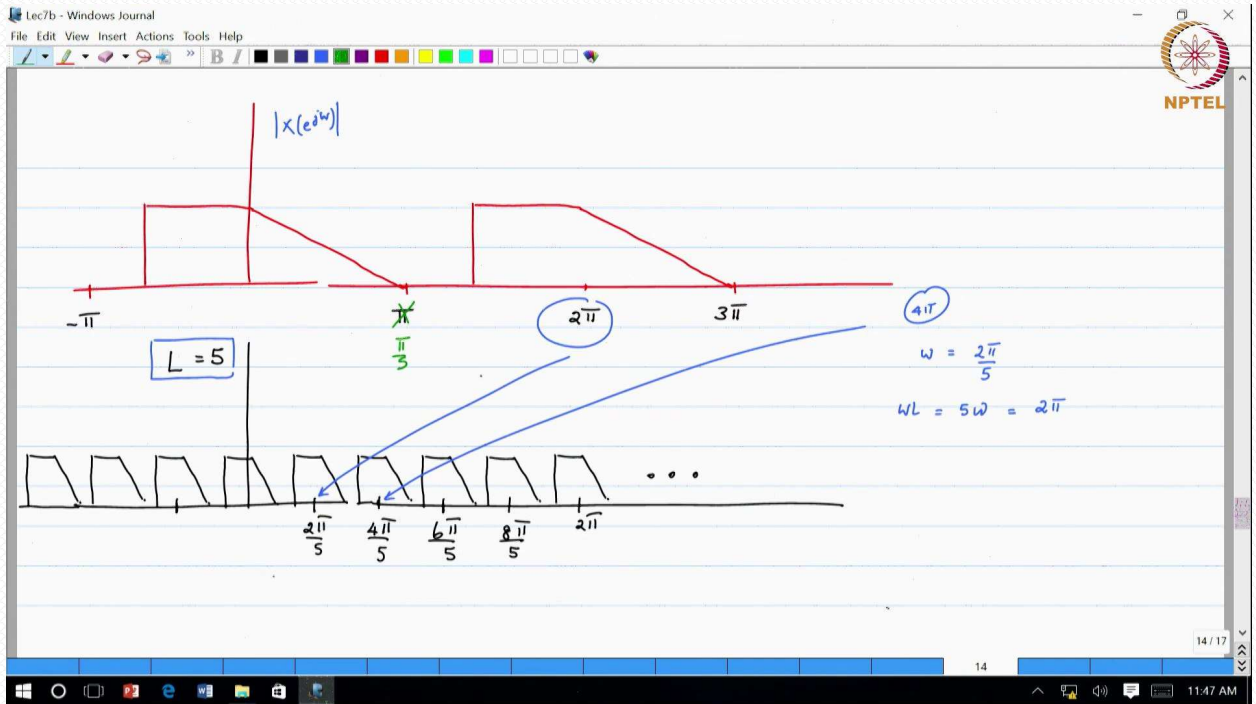
$y_L[n] = x\left[\frac{n}{L}\right]$

$\rightarrow (L-1)$  zeros between samples

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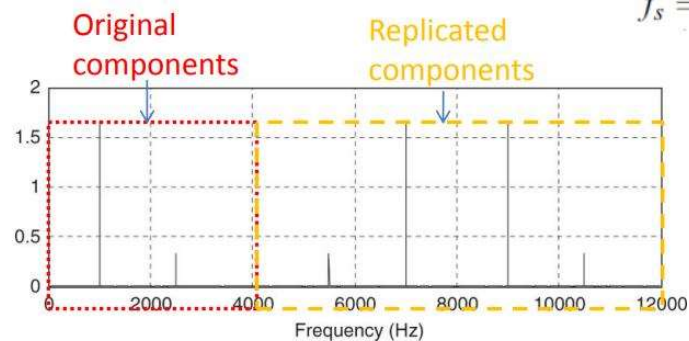
# Upsampling (Interpolation)



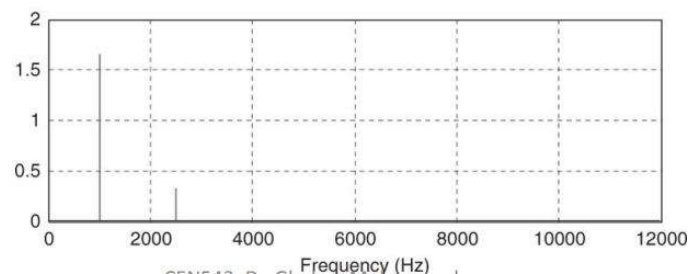
# Upsampling (Interpolation)

$$x(n) = 5 \sin\left(\frac{2\pi \times 1000n}{8000}\right) + \cos\left(\frac{2\pi \times 2500n}{8000}\right)$$

$$f_s = 8,000 \text{ Hz}$$



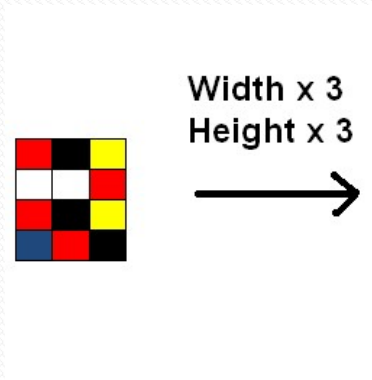
After applying  
interpolation  
filter



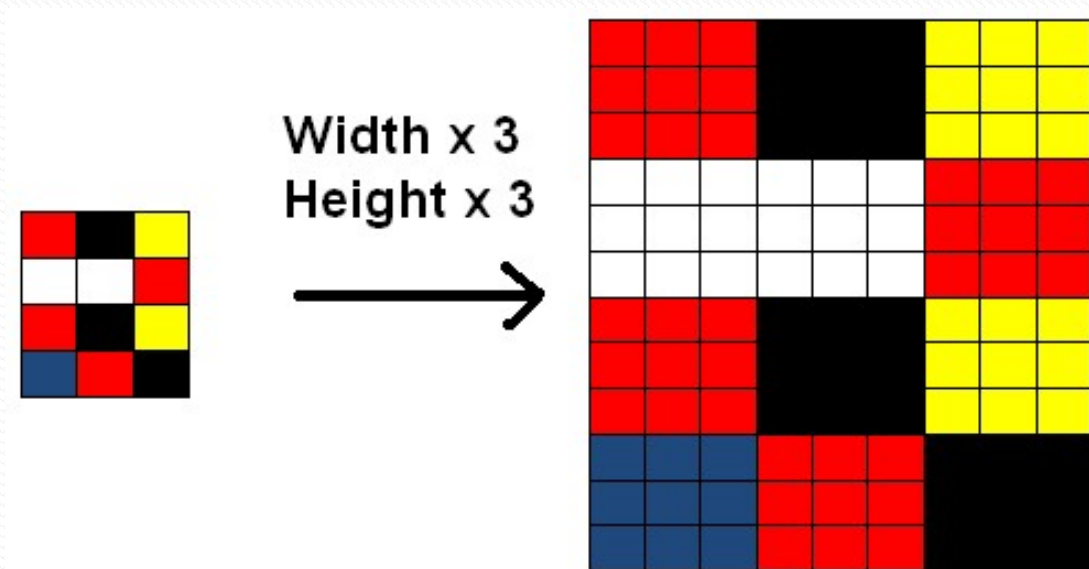




# Upsampling (Interpolation)



# Upsampling (Interpolation)

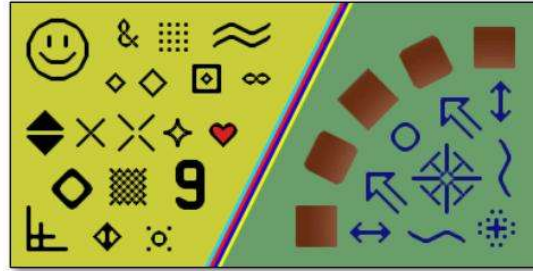




# Upsampling (Interpolation)



Nearest-neighbor interpolation



hq3x interpolation (ZSNES) <http://en.wikipedia.org/wiki/Hqx>



# Downsampling (Decimation)

Input:  $x(n)$ . Output:  $y(m) = x(mM)$ .



Taking one sample from every  $M$  samples in the input.

**Example:**  $x(n)$ : 8 7 4 8 9 6 4 2 -2 -5 -7 -7 -6 -4...



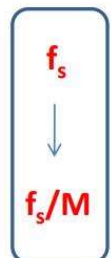
Downsample by a factor of 3

$y(m)$ : 8 8 4 -5 -6...

Let, sampling period of  $x(n)$  is 0.1 second [10 samples per second]



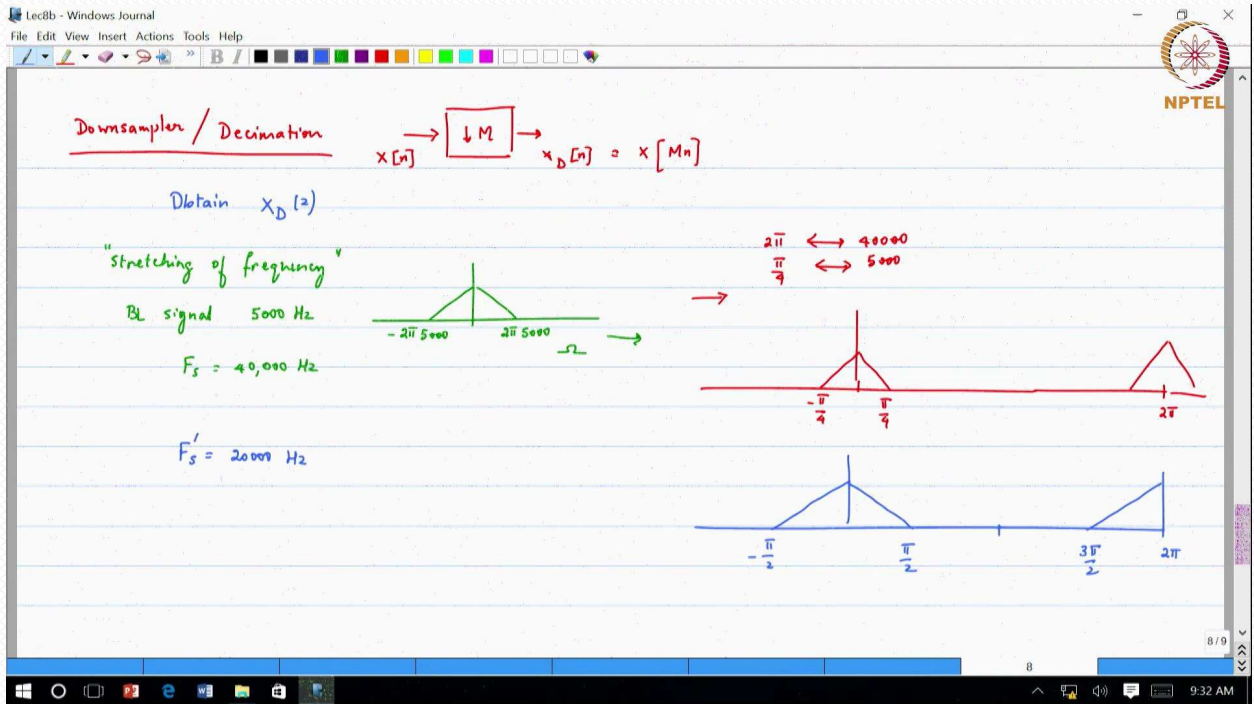
Sampling period of  $y(m)$  is  $3 \times 0.1$  second [3.3 samples per second]



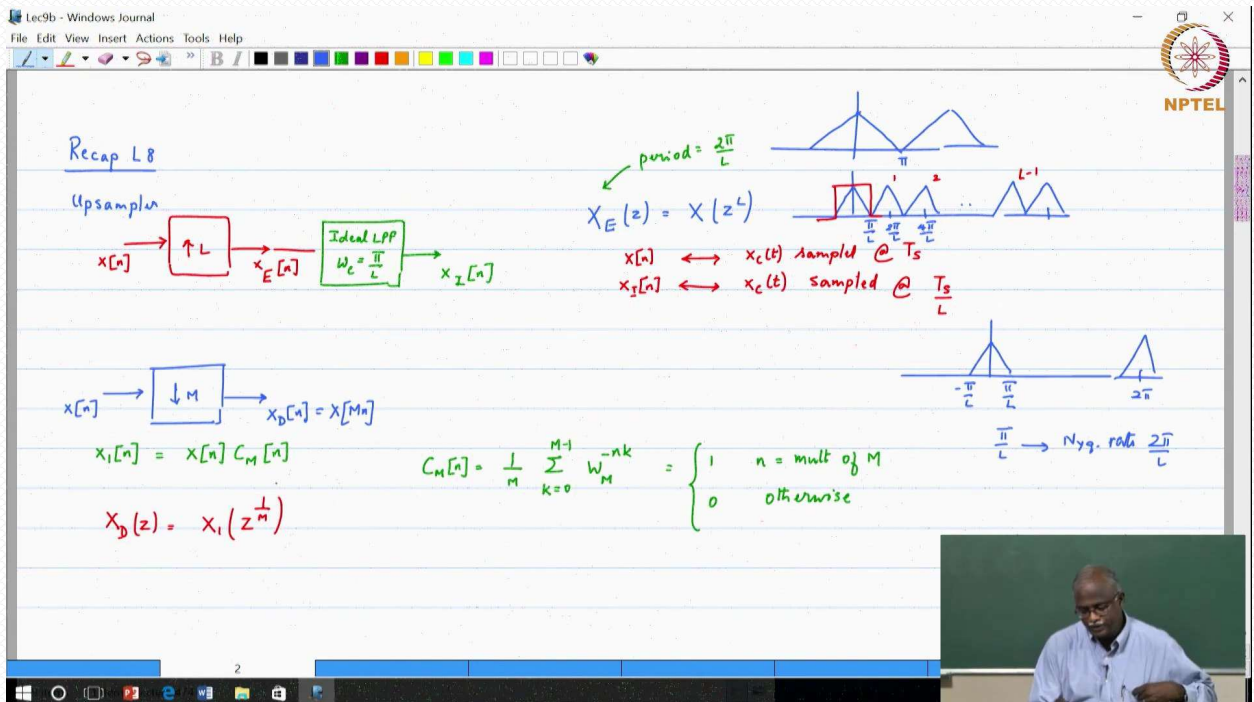




# Downsampling (Decimation)



# Downsampling (Decimation)



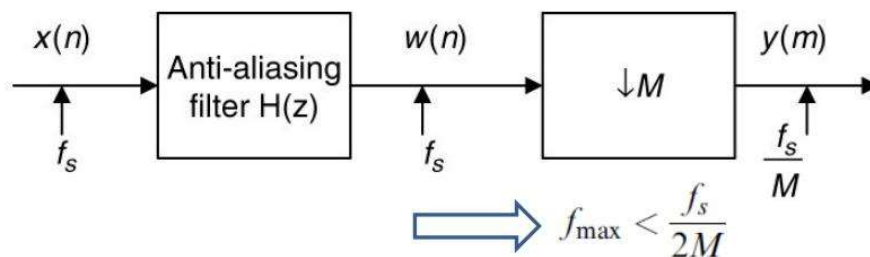


# Downsampling (Decimation)

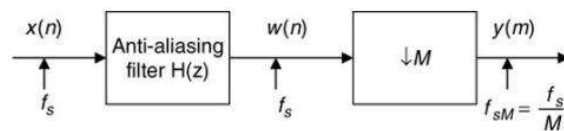
Folding frequency after downsampling:  $f_{sM}/2 = \frac{f_s}{2M}$

If the signal to be downsampled has frequency components larger than the folding frequency, aliasing noise will be introduced into the downsampled data.

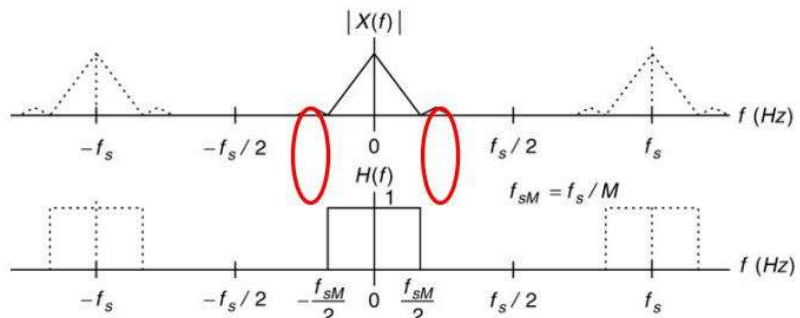
Anti-aliasing filter:  $H(z)$  → Low pass filter with stop frequency edge at  $f_{sM}/2 = \frac{f_s}{2M}$



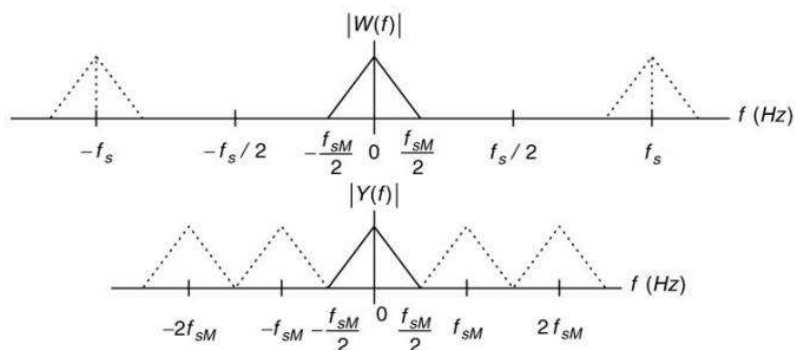
# Downsampling (Decimation)



Frequency response of anti-aliasing filter



Output spectrum



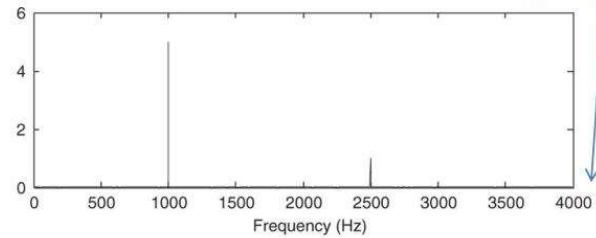




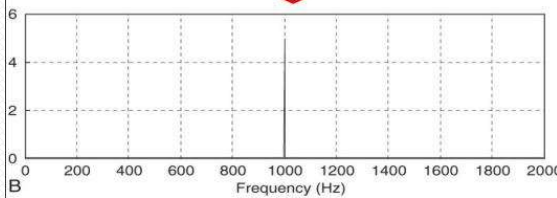
# Downsampling (Decimation)

Let:  $x(n) = 5 \sin\left(\frac{2\pi \times 1000n}{8000}\right) + \cos\left(\frac{2\pi \times 2500n}{8000}\right)$   $f_s = 8,000$  Hz

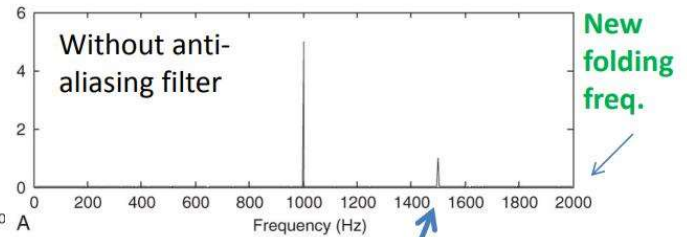
Original  
folding  
freq.



With anti-aliasing filter



Without anti-aliasing filter



New  
folding  
freq.

$$4 \text{ kHz} - 2.5 \text{ kHz} = 1.5 \text{ kHz}^5$$